Kemal Ozturk

SOFTWARE ENGINEER · PRODUCT DEVELOPMENT & SYSTEMS DESIGN

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PROFESSIONAL SUMMARY

Software engineer with 7 years of experience designing, shipping, and operating software across web, desktop, and backend services. Leads projects from technical design through production rollout, with experience in reusable architecture, authorization, concurrency, and reliability. Maintains open-source software spanning Rust services, Linux/Windows hardware integration, and web applications.

TECHNICAL SKILLS

Product: TypeScript, JavaScript, React, Angular, Next.js, Svelte, Redux/Redux-Saga, Node.js, HTML, CSS/SCSS, Tailwind

Platform: REST APIs, Postgres, Firebase, GCP, AWS, S3, Temporal, Cloudflare, Sentry, Grafana, LaunchDarkly

Systems: Rust, Tokio, Poem/OpenAPI, Python, C, C++, Linux, Nix/NixOS, Arch Linux

Media & real time: WebRTC, Daily, MediaRecorder, WebCodecs, WebAssembly, Web Workers, Electron

Development tools: GitHub Actions, Nx, Turborepo, Vite, Retool, Figma, Cursor, Claude Code, OpenAI Codex, Zed

PROFESSIONAL EXPERIENCE

Software Engineer

June 2023 - September 2026

Descript

- Architected the Web Recorder engine with Redux-Saga as a reusable TypeScript package. Editor, Media Library, Quick Recorder, and Rooms were built on top of the engine. It supported multi-track recording and device switching. Inputs included 4K/30 FPS camera video, microphone and system audio, and screen capture.
- Led Scheduled Rooms from technical design through phased rollout, coordinating work across APIs, permissions, and recording clients. The feature reached **500+ creations per week** and roughly **25% of Rooms recordings**.
- Designed guest access for Scheduled Rooms so invitees could record without joining or accessing the host's project. Enforced tightly scoped permissions on the server. Coordinated the security review and staged rollout to general availability.
- Built a Web Worker upload manager with upload chunking and progress-aware timeouts. Fixed an asynchronous race in the segment counter logic to improve upload reliability.
- Owned and shipped recording-script visualization, recording cancel and restart, and pre-roll/live playback.
- Added an internal denoise audio effect using an RNNoise-backed WebAssembly package.
- Reduced the drift between audio and screen video capture from about **700 ms to 50 ms** by profiling capture timings and adding delay-compensation logic.
- Led response to recording failures across web and desktop. Investigated recovery paths, device failures, authorization errors, and rate limits. Built dashboards, alerts, browser-memory telemetry, preflight deployment checks, and tools that helped Support diagnose recording failures.
- Volunteered as the engineering contact for an at-risk enterprise trial. Shipped 2 fixes and joined a customer meeting with Product, Support, and Marketing. The customer converted after the team addressed nearly all concerns.

Software Engineer
January 2022 - August 2023

SquadCast (acquired by Descript)

- Built remote-recording features in Angular, TypeScript, Next.js, Firebase, and GCP across capture, authentication, account management, third-party integrations, and onboarding.
- Migrated selected microservices to serverless cloud functions to reduce costs and support scaling.
- Designed and shipped Canvas and SquadShots. Added app-wide dark mode, mobile-first layouts, and accessibility improvements.
- Profiled CPU use and fixed memory leaks across Studio, Chat, Dashboard, and audio metering.
- Cut new-feature scaffolding time from **30 minutes to 5** by building Nx executors and generators that standardized setup.

Software Engineer
August 2019 - September 2021

Peel9

- Delivered records-management features used by **10+ police departments** around Cincinnati, including incident approval, traffic-crash diagrams, modular forms, and account recovery, using Node.js, Express, and Handlebars.
- Led code reviews, mentored new hires, and maintained the GitHub repository.
- Built a Leaflet plugin for dynamic crash-map markers.

SELECTED PROJECTS

Framework Control
Volunteer/open source - 2026-Present

Built and maintain a Rust service with Tokio and Poem/OpenAPI plus a Svelte UI for Framework laptops and desktops on Linux and Windows. Added telemetry, rate-limited fan curves with hysteresis, battery/power controls, a Windows MSI installer, Linux systemd and udev setup, and packages for AUR and nixpkgs. Added tests for fan-control behavior and CLI parsing. The project has **225+ GitHub stars and 7+ forks**.

Cihan Radyo
Volunteer - Ongoing

Built and maintain a Svelte/TypeScript radio and podcast PWA with local playback state, lock-screen controls, Turkish-aware search, Google Drive sync, and a publishing pipeline for podcast RSS feeds. Released on Google Play and the iOS App Store. Recent Cloudflare data shows **1.9K unique visitors, 334K requests, and 20 GB served daily**.

RGBPiano
Personal project - 2023-Present

Built a MIDI-driven piano LED visualizer. A TypeScript host computes note mapping and envelopes into LED frames and sends them over WebSocket to a Python service on a Raspberry Pi Zero that drives WS2812 LEDs. A Svelte UI previews and configures the visual behavior.

EDUCATION

Bachelor of Science, Information Technology
2021 - GPA 3.97/4.0

University of Cincinnati

Coursework toward a Bachelor of Science in Computer Science
2016-2018 - GPA 3.83/4.0

Ankara University

Transferred to the University of Cincinnati; no degree was conferred by Ankara University.